

Huamin Wang

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RESEARCH LEADERSHIP AND APPOINTMENTS

2022–Present	Chief Scientist, Style3D (Lintex Digital Inc.) Lead Style3D's global research team and shape its research agenda across computational physics, learned physical models, and embodied intelligence. Align research priorities with company strategy and work with product teams to translate research into digital-fashion capabilities and emerging robotics applications.
2017–2022	Associate Professor, tenured, The Ohio State University Department of Computer Science and Engineering
2011–2017	Assistant Professor, The Ohio State University Department of Computer Science and Engineering
2009–2011	Postdoctoral Researcher, University of California, Berkeley Advisors: James F. O'Brien and Ravi Ramamoorthi

RESEARCH FOCUS

My research advances the computational foundations of physical intelligence: simulating physical systems, learning their behavior from real-world data, and enabling robots to interact with them.

- **Computational Physics.** Accurate, robust, and scalable simulation of rigid and deformable systems. My work emphasizes numerical optimization, GPU acceleration, and strongly coupled contact and interaction, providing efficient foundations for complex physical environments.
- **Learning Physical Dynamics.** Neural and hybrid simulators that learn physical behavior from data, together with real-to-sim methods for material characterization, parameter identification, and model calibration. I combine physical priors with measured behavior to improve predictive fidelity and generalization.
- **Embodied Intelligence.** World models and interactive simulators for physical AI and robotics, including robot learning, manipulation, and evaluation. I investigate how generated environments and real-world interaction can expand training experience and support continual improvement of models and policies.

SELECTED RESEARCH CONTRIBUTIONS

Efficient physical computation. Developed and co-developed GPU cloth simulation, nonlinear multigrid, and domain-decomposed projective dynamics methods, with work recognized by the Pacific Graphics 2018 Best Paper Award and a SIGGRAPH 2025 Best Paper Honorable Mention.

Physical models from data. Contributions span measured elastic models for cloth, elastic parameter optimization, learning-based bending stiffness estimation, neural deformation, and neural-assisted cloth homogenization, connecting real-world measurements with predictive simulation.

Interactive worlds for intelligent agents. Co-developed RGBench for robotic garment manipulation and structured garment generation methods including GarmageNet and Design2GarmentCode. Current research extends this foundation toward world models and robot–object interaction.

EDUCATION

2004–2009	Ph.D. in Computer Science , Georgia Institute of Technology Advisor: Greg Turk
2003–2004	M.S. in Computer Science , Stanford University Advisor: Leo Guibas
1998–2002	B.Eng. in Computer Science and Engineering , Zhejiang University Highest honors; Mixed Class '98 (honors program).

HONORS AND RECOGNITION

2026	Springer Nature Editor of Distinction Award
2025	ACM Distinguished Member IEEE Senior Member Best Paper Honorable Mention ACM SIGGRAPH 2025 High-Performance CPU Cloth Simulation Using Domain-Decomposed Projective Dynamics
2018	Best Paper Award Pacific Graphics Parallel Multigrid for Nonlinear Cloth Simulation
2017	Lumley Research Award The Ohio State University
2013–2018	Research Gift NVIDIA Corporation
2012–2018	Research Gift Adobe Inc.
2006	Graduate Fellowship NVIDIA Corporation

PROFESSIONAL SERVICE

Editorial Appointments

2024–Present	Associate Editor IEEE Transactions on Visualization and Computer Graphics (TVCG)
2020–Present	Associate Editor The Visual Computer

Award Committees

2026–2029	Committee Member ACM Doctoral Dissertation Award
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Conference Leadership

2026	Assistant Technical Papers Chair ACM SIGGRAPH Asia
2025	Steering Committee Member ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA)
2023	Technical Program Chair ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA)
2017	Technical Program Chair Computer Animation and Social Agents (CASA)

Program Committees

2014, 2015, 2018–2025	ACM SIGGRAPH, ACM SIGGRAPH Asia
2010, 2014–2017	Pacific Graphics
2012–2014, 2018	ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)
2012, 2013	Computer Animation and Social Agents (CASA)
2012–2018	ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA)
2013, 2017	Computer Aided Design and Computer Graphics (CAD & Graphics)

Reviewing and Panels

2005–Present. National Science Foundation (NSF); ACM SIGGRAPH; ACM SIGGRAPH Asia; Eurographics; Computer Graphics International (CGI); Eurographics Symposium on Rendering (EGSR); SCA; I3D; Computer Graphics Forum (CGF); CASA; ACM Transactions on Graphics (TOG); IEEE TVCG; CAD & Graphics; CVPR; ICCV; ECCV.

JOURNAL AND CONFERENCE PUBLICATIONS

Reverse chronological order by year and venue date; papers from the same conference edition are grouped.

- 2026 **Multi-level quadrature for projective dynamics.**
Dewen Guo, Yuqi Meng, Zixuan Lu, Zhiyong He, **Huamin Wang**, Lei Lan, Weiwei Xu, Kui Wu, Chenfanfu Jiang, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH Asia 2026 journal track), 2026. Conditionally accepted.
- Automatic sewing pattern layout by graph-based two-level semantic classification.**
Zhendong Wang, Haofeng Zhu, Chen Liu, Yao Jin, Ligang Liu, and **Huamin Wang**.
ACM SIGGRAPH Asia Conference Papers, 2026. Conditionally accepted.
- TailorCoPilot: Enabling agentic pattern making with version-controlled state tracking.**
Yuxin Sun, Zhaohui Wang, Ruiyang Liu, Demian Kong, Qian He, Gaofeng He, and **Huamin Wang**.
ACM Symposium on User Interface Software and Technology (UIST), 2026.
- Interactive yarn-level knitwear with nested Douglas-Rachford splitting.**
Chun Yuan, Zixuan Lu, Haoyang Shi, Dewen Guo, **Huamin Wang**, Chenfanfu Jiang, Zherong Pan, Kui Wu, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH), vol. 45, no. 4, 2026.
- Heterogeneous subspace corrections for GPU deformable multibody dynamics.**
Dewen Guo, Zhendong Wang, Minchen Li, Sheng Li, Guoping Wang, **Huamin Wang**, Chenfanfu Jiang, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH), vol. 45, no. 4, 2026.
- FabricTryOn: Taming image editing models for garment re-texturing.**
Jun Ma, Qian He, Gaofeng He, Huang Chen, Chen Liu, Xiaogang Jin, Yin Yang, and **Huamin Wang**.
ACM SIGGRAPH Conference Papers, 2026.
- Distributed affine body dynamics with adaptive consensus.**
Jiafeng Liu, Wenhui Zhou, Xinming Pei, Yifan Peng, **Huamin Wang**, Yin Yang, Lei Lan, and Weiwei Xu.
ACM SIGGRAPH Conference Papers, 2026.
- CAN: A curvature-aware Nesterov optimizer for fast elastic simulation with topological changes.**
Yuxiong Qin, **Huamin Wang**, Qingfu Zhang, and Zhongkai Zhang.
IEEE Transactions on Visualization and Computer Graphics, pp. 1–14, 2026.
- Real Garment Benchmark (RGBench): A comprehensive benchmark for robotic garment manipulation featuring a high-fidelity scalable simulator.**
Wenkang Hu, Xincheng Tang, Yanzhi E, Yitong Li, Zhengjie Shu, Wei Li, **Huamin Wang**, and Ruigang Yang.
Proceedings of the AAAI Conference on Artificial Intelligence, vol. 40, no. 22, pp. 18306–18314, 2026.
- 2025 **Progressive outfit assembly and instantaneous pose transfer.**
Dewen Guo, Zhendong Wang, Zegao Liu, Sheng Li, Guoping Wang, Yin Yang, and **Huamin Wang**.
ACM SIGGRAPH Asia Conference Papers, pp. 1–12, 2025.
- One-shot embroidery customization via contrastive LoRA modulation.**
Jun Ma, Qian He, Gaofeng He, Huang Chen, Chen Liu, Xiaogang Jin, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 44, no. 6, Article 271, 15 pp., 2025.
- GarmageNet: A multimodal generative framework for sewing pattern design and generic garment modeling.**
Siran Li, Ruiyang Liu, Chen Liu, Zhendong Wang, Gaofeng He, Yong-Lu Li, Xiaogang Jin, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 44, no. 6, Article 216, 23 pp., 2025.
- Efficient object reconstruction with differentiable area light shading.**
Yaoan Gao, Jiamin Xu, James Tompkin, Qi Wang, Zheng Dong, Hujun Bao, Yujun Shen, **Huamin Wang**, Changqing Zou, and Weiwei Xu.
ACM SIGGRAPH Asia Conference Papers, pp. 1–12, 2025.
- Physics-inspired estimation of optimal cloth mesh resolution.**
Diyang Zhang, Zhendong Wang, Zegao Liu, Xinming Pei, Weiwei Xu, and **Huamin Wang**.
ACM SIGGRAPH Conference Papers, Article 23, 11 pp., 2025.

2025

JGS2: Near second-order converging Jacobi/Gauss-Seidel for GPU elastodynamics.

Lei Lan, Zixuan Lu, Chun Yuan, Weiwei Xu, Hao Su, **Huamin Wang**, Chenfanfu Jiang, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH), vol. 44, no. 4, Article 44, 15 pp., 2025.

High-performance CPU cloth simulation using domain-decomposed projective dynamics.

Zixuan Lu, Ziheng Liu, Lei Lan, **Huamin Wang**, Yuko Ishiwaka, Chenfanfu Jiang, Kui Wu, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH), vol. 44, no. 4, Article 51, 17 pp., 2025.

Fast physics-based modeling of knots and ties using templates.

Dewen Guo, Zhendong Wang, Zegao Liu, Sheng Li, Guoping Wang, Yin Yang, and **Huamin Wang**.
ACM SIGGRAPH Conference Papers, Article 43, 9 pp., 2025.

Automated task scheduling for cloth and deformable body simulations in heterogeneous computing environments.

Chengzhu He, Zhendong Wang, Zhaorui Meng, Junfeng Yao, Shihui Guo, and **Huamin Wang**.
ACM SIGGRAPH Conference Papers, Article 24, 11 pp., 2025.

Design2GarmentCode: Turning design concepts to tangible garments through program synthesis.

Feng Zhou, Ruiyang Liu, Chen Liu, Gaofeng He, Yong-Lu Li, Xiaogang Jin, and **Huamin Wang**.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 23712–23722, 2025.

GarmentDreamer: 3DGS guided garment synthesis with diverse geometry and texture details.

Boqian Li, Xuan Li, Ying Jiang, Tianyi Xie, Feng Gao, **Huamin Wang**, Yin Yang, and Chenfanfu Jiang.
International Conference on 3D Vision (3DV), pp. 1416–1426, 2025.

2024

FashionR2R: Texture-preserving rendered-to-real image translation with diffusion models.

Rui Hu, Qian He, Gaofeng He, Jiedong Zhuang, Huang Chen, Huafeng Liu, and **Huamin Wang**.
Advances in Neural Information Processing Systems (NeurIPS), Article 1107, 30 pp., 2024.

Volumetric homogenization for knitwear simulation.

Chun Yuan, Haoyang Shi, Lei Lan, Yuxing Qiu, Cem Yuksel, **Huamin Wang**, Chenfanfu Jiang, Kui Wu, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 43, no. 6, Article 207, 19 pp., 2024.

Efficient GPU cloth simulation with non-distance barriers and subspace reuse.

Lei Lan, Zixuan Lu, Jingyi Long, Chun Yuan, Xuan Li, Xiaowei He, **Huamin Wang**, Chenfanfu Jiang, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 43, no. 6, Article 226, 16 pp., 2024.

Projective peridynamic modeling of hyperelastic membranes with contact.

Zixuan Lu, Xiaowei He, Yuzhong Guo, Xuehui Liu, and **Huamin Wang**.
IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 8, pp. 4601–4614, 2024.

VR-GS: A physical dynamics-aware interactive Gaussian splatting system in virtual reality.

Ying Jiang, Chang Yu, Tianyi Xie, Xuan Li, Yutao Feng, **Huamin Wang**, Minchen Li, Henry Lau, Feng Gao, Yin Yang, and Chenfanfu Jiang.
ACM SIGGRAPH Conference Papers, Article 78, 1 pp., 2024.

Neural-assisted homogenization of yarn-level cloth.

Xudong Feng, **Huamin Wang**, Yin Yang, and Weiwei Xu.
ACM SIGGRAPH Conference Papers, Article 80, 10 pp., 2024.

High-quality surface reconstruction using Gaussian surfels.

Pinxuan Dai, Jiamin Xu, Wenxiang Xie, Xinguo Liu, **Huamin Wang**, and Weiwei Xu.
ACM SIGGRAPH Conference Papers, Article 22, 11 pp., 2024.

A dynamic duo of finite elements and material points.

Xuan Li, Minchen Li, Xuchen Han, **Huamin Wang**, Yin Yang, and Chenfanfu Jiang.
ACM SIGGRAPH Conference Papers, Article 97, 11 pp., 2024.

Automatic digital garment initialization from sewing patterns.

Chen Liu, Weiwei Xu, Yin Yang, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH), vol. 43, no. 4, Article 74, 12 pp., 2024.

2023

Subspace-preconditioned GPU projective dynamics with contact for cloth simulation.

Xuan Li, Yu Fang, Lei Lan, **Huamin Wang**, Yin Yang, Minchen Li, and Chenfanfu Jiang.
ACM SIGGRAPH Asia Conference Papers, Article 1, 12 pp., 2023.

- 2023 **Stable discrete bending by analytic eigensystem and adaptive orthotropic geometric stiffness.**
Zhendong Wang, Yin Yang, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 42, no. 6, Article 175, 16 pp., 2023.
- Second-order stencil descent for interior-point hyperelasticity.**
Lei Lan, Minchen Li, Chenfanfu Jiang, **Huamin Wang**, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH), vol. 42, no. 4, Article 108, 16 pp., 2023.
- Fast GPU-based two-way continuous collision handling.**
Tianyu Wang, Jiong Chen, Dongping Li, Xiaowei Liu, **Huamin Wang**, and Kun Zhou.
ACM Transactions on Graphics (SIGGRAPH), vol. 42, no. 5, Article 167, 15 pp., 2023.
- 2022 **Learning-based bending stiffness parameter estimation by a drape tester.**
Xudong Feng, Wenchao Huang, Weiwei Xu, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 41, no. 6, Article 221, 16 pp., 2022.
- A GPU-based multilevel additive Schwarz preconditioner for cloth and deformable body simulation.**
Botao Wu, Zhendong Wang, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH), vol. 41, no. 4, Article 63, 14 pp., 2022.
- 2021 **GPU-based simulation of cloth wrinkles at submillimeter levels.**
Huamin Wang.
ACM Transactions on Graphics (SIGGRAPH), vol. 40, no. 4, Article 169, 14 pp., 2021.
- Computational design of skinned quad-robots.**
Xudong Feng, Jiafeng Liu, **Huamin Wang**, Yin Yang, Hujun Bao, Bernd Bickel, and Weiwei Xu.
IEEE Transactions on Visualization and Computer Graphics, vol. 27, no. 6, pp. 2881–2895, 2021.
- 2020 **Interactive liquid splash modeling by user sketches.**
Guowei Yan, Zhili Chen, Jimei Yang, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 39, no. 6, Article 165, 13 pp., 2020.
- A safe and fast repulsion method for GPU-based cloth self collisions.**
Longhua Wu, Botao Wu, Yin Yang, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH), vol. 40, no. 1, Article 5, 18 pp., 2020.
- Medial elastics: Efficient and collision-ready deformation via medial axis transform.**
Lei Lan, Ran Luo, Marco Fratarcangeli, Weiwei Xu, **Huamin Wang**, Xiaohu Guo, Junfeng Yao, and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH), vol. 39, no. 3, Article 20, 17 pp., 2020.
- NNWarp: Neural network-based nonlinear deformation.**
Ran Luo, Tianjia Shao, **Huamin Wang**, Weiwei Xu, Xiang Chen, Kun Zhou, and Yin Yang.
IEEE Transactions on Visualization and Computer Graphics, vol. 26, no. 4, pp. 1745–1759, 2020.
- 2018 **Parallel iterative solvers for real-time elastic deformations.**
Marco Fratarcangeli, **Huamin Wang**, and Yin Yang.
SIGGRAPH Asia Courses, Article 14, 45 pp., 2018.
- Inexact descent methods for elastic parameter optimization.**
Guowei Yan, Wei Li, Ruigang Yang, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 37, no. 6, Article 253, 14 pp., 2018.
- Physics-based quadratic deformation using elastic weighting.**
Ran Luo, Weiwei Xu, **Huamin Wang**, Kun Zhou, and Yin Yang.
IEEE Transactions on Visualization and Computer Graphics, vol. 24, no. 12, pp. 3188–3199, 2018.
- Parallel multigrid for nonlinear cloth simulation.**
Zhendong Wang, Longhua Wu, Marco Fratarcangeli, Min Tang, and **Huamin Wang**.
Computer Graphics Forum (Pacific Graphics), vol. 37, no. 7, pp. 131–141, 2018.
- Projective peridynamics for modeling versatile elastoplastic materials.**
Xiaowei He, **Huamin Wang**, and Enhua Wu.
IEEE Transactions on Visualization and Computer Graphics, vol. 24, no. 9, pp. 2589–2599, 2018.

- 2018 **Rule-free sewing pattern adjustment with precision and efficiency.**
Huamin Wang.
ACM Transactions on Graphics (SIGGRAPH), vol. 37, no. 4, Article 53, 13 pp., 2018.
- Interactive two-way shape design of elastic bodies.**
 Rajaditya Mukherjee, Longhua Wu, and **Huamin Wang.**
Proceedings of the ACM on Computer Graphics and Interactive Techniques (I3D), vol. 1, no. 1, Article 11, 17 pp., 2018.
- 2017 **Modeling, evaluation and optimization of interlocking shell pieces.**
 Miaojun Yao, Zhili Chen, Weiwei Xu, and **Huamin Wang.**
Computer Graphics Forum (Pacific Graphics), vol. 36, no. 7, pp. 1–13, 2017.
- SPH-based simulation of liquid wetting across textile materials.**
 Aihua Mao, Mingle Wang, Yong-Jin Liu, **Huamin Wang**, and Guiqing Li.
Communications in Information and Systems, vol. 17, no. 3, pp. 147–169, 2017.
- 2016 **Descent methods for elastic body simulation on the GPU.**
Huamin Wang and Yin Yang.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 35, no. 6, Article 212, 10 pp., 2016.
- Incremental deformation subspace reconstruction.**
 Rajaditya Mukherjee, Xiaofeng Wu, and **Huamin Wang.**
Computer Graphics Forum (Pacific Graphics), vol. 35, no. 7, pp. 169–178, 2016.
- Enriching SPH simulation by approximate capillary waves.**
 Sheng Yang, Xiaowei He, **Huamin Wang**, Sheng Li, Guoping Wang, Enhua Wu, and Kun Zhou.
ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA), pp. 29–36, 2016.
- CAMA: Contact-aware matrix assembly with unified collision handling for GPU-based cloth simulation.**
 Min Tang, **Huamin Wang**, Le Tang, Ruofeng Tong, and Dinesh Manocha.
Computer Graphics Forum (Eurographics), vol. 35, no. 2, pp. 511–521, 2016.
- 2015 **Wetbrush: GPU-based 3D painting simulation at the bristle level.**
 Zhili Chen, Byungmoon Kim, Daichi Ito, and **Huamin Wang.**
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 34, no. 6, Article 200, 11 pp., 2015.
- A unified approach for subspace simulation of deformable bodies in multiple domains.**
 Xiaofeng Wu, Rajaditya Mukherjee, and **Huamin Wang.**
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 34, no. 6, Article 241, 9 pp., 2015.
- Level-set-based partitioning and packing optimization of a printable model.**
 Miaojun Yao, Zhili Chen, Linjie Luo, Rui Wang, and **Huamin Wang.**
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 34, no. 6, Article 214, 11 pp., 2015.
- A Chebyshev semi-iterative approach for accelerating projective and position-based dynamics.**
Huamin Wang.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 34, no. 6, Article 246, 9 pp., 2015.
- Simulation of fluid mixing with interface control.**
 Xiaowei He, **Huamin Wang**, Fengjun Zhang, Hongan Wang, Guoping Wang, Kun Zhou, and Enhua Wu.
ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA), pp. 129–135, 2015.
- Robust simulation of sparsely sampled thin features in SPH-based free surface flows.**
 Xiaowei He, **Huamin Wang**, Fengjun Zhang, Hongan Wang, Guoping Wang, and Kun Zhou.
ACM Transactions on Graphics (SIGGRAPH), vol. 34, no. 1, Article 7, 9 pp., 2015.
- Automatic posing of a meshed human model using point clouds.**
 Tamal K. Dey, Bo Fu, **Huamin Wang**, and Lei Wang.
Computers & Graphics (Shape Modeling International), vol. 46, pp. 14–24, 2015.
- 2014 **Physics-inspired adaptive fracture refinement.**
 Zhili Chen, Miaojun Yao, Renguo Feng, and **Huamin Wang.**
ACM Transactions on Graphics (SIGGRAPH), vol. 33, no. 4, Article 113, 7 pp., 2014.

- 2014 **Defending continuous collision detection against errors.**
Huamin Wang.
ACM Transactions on Graphics (SIGGRAPH), vol. 33, no. 4, Article 122, 10 pp., 2014.
- Real-time human pose and shape estimation for virtual try-on using a single commodity depth camera.**
 Mao Ye, **Huamin Wang**, Nianchen Deng, Xubo Yang, and Ruigang Yang.
IEEE Transactions on Visualization and Computer Graphics (IEEE VR), vol. 20, no. 4, pp. 550–559, 2014.
- 2013 **Bilateral blue noise sampling.**
 Jiating Chen, Xiaoyin Ge, Li-Yi Wei, Bin Wang, Yusu Wang, **Huamin Wang**, Yun Fei, Kang-Lai Qian, Jun-Hai Yong, and Wenping Wang.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 32, no. 6, Article 216, 11 pp., 2013.
- Modeling friction and air effects between cloth and deformable bodies.**
 Zhili Chen, Renguo Feng, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH), vol. 32, no. 4, Article 88, 8 pp., 2013.
- Adaptive fracture simulation of multi-layered thin plates.**
 Oleksiy Busaryev, Tamal K. Dey, and **Huamin Wang**.
ACM Transactions on Graphics (SIGGRAPH), vol. 32, no. 4, Article 52, 6 pp., 2013.
- 2012 **Simulation guided hair dynamics modeling from video.**
 Qing Zhang, Jing Tong, **Huamin Wang**, Zhigeng Pan, and Ruigang Yang.
Computer Graphics Forum (Pacific Graphics), vol. 31, no. 7, pp. 2003–2010, 2012.
- Data-driven simulation methods in computer graphics: Cloth, tissue and faces.**
 Miguel A. Otaduy, Bernd Bickel, Derek Bradley, and **Huamin Wang**.
ACM SIGGRAPH Courses, Article 12, 96 pp., 2012.
- Animating bubble interactions in a liquid foam.**
 Oleksiy Busaryev, Tamal K. Dey, **Huamin Wang**, and Zhong Ren.
ACM Transactions on Graphics (SIGGRAPH), vol. 31, no. 4, Article 63, 8 pp., 2012.
- A deformable surface model for real-time water drop animation.**
 Yizhong Zhang, **Huamin Wang**, Shuai Wang, Yiyong Tong, and Kun Zhou.
IEEE Transactions on Visualization and Computer Graphics, vol. 18, no. 8, pp. 1281–1289, 2012.
- Compressing repeated content within large-scale remote sensing images.**
 Wei Hua, Rui Wang, Xusheng Zeng, Ying Tang, **Huamin Wang**, and Hujun Bao.
The Visual Computer (Computer Graphics International), vol. 28, no. 6-8, pp. 755–764, 2012.
- 2011 **Data-driven elastic models for cloth: Modeling and measurement.**
Huamin Wang, James F. O'Brien, and Ravi Ramamoorthi.
ACM Transactions on Graphics (SIGGRAPH), vol. 30, no. 4, Article 71, 12 pp., 2011.
- 2010 **Multi-resolution isotropic strain limiting.**
Huamin Wang, James O'Brien, and Ravi Ramamoorthi.
ACM Transactions on Graphics (SIGGRAPH Asia), vol. 29, no. 6, Article 156, 10 pp., 2010.
- Example-based wrinkle synthesis for clothing animation.**
Huamin Wang, Florian Hecht, Ravi Ramamoorthi, and James F. O'Brien.
ACM Transactions on Graphics (SIGGRAPH), vol. 29, no. 4, Article 107, 8 pp., 2010.
- 2009 **Modeling deformable objects from a single depth camera.**
 Miao Liao, Qing Zhang, **Huamin Wang**, Ruigang Yang, and Minglun Gong.
IEEE International Conference on Computer Vision (ICCV), pp. 167–174, 2009.
- Material space texturing.**
 Nicolas Ray, Bruno Lévy, **Huamin Wang**, Greg Turk, and Bruno Vallet.
Computer Graphics Forum, vol. 28, no. 6, pp. 1659–1669, 2009.
- Physically guided liquid surface modeling from videos.**
Huamin Wang, Miao Liao, Qing Zhang, Ruigang Yang, and Greg Turk.
ACM Transactions on Graphics (SIGGRAPH), vol. 28, no. 3, Article 90, 11 pp., 2009.

- 2008 **Factoring repeated content within and among images.**
Huamin Wang, Yonatan Wexler, Eyal Ofek, and Hugues Hoppe.
ACM Transactions on Graphics (SIGGRAPH), vol. 27, no. 3, pp. 1–10, 2008.
- 2007 **Solving general shallow wave equations on surfaces.**
Huamin Wang, Gavin Miller, and Greg Turk.
ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA), pp. 229–238, 2007.
- Space-time light field rendering.**
Huamin Wang, Mingxuan Sun, and Ruigang Yang.
IEEE Transactions on Visualization and Computer Graphics, vol. 13, no. 4, pp. 697–710, 2007.
- 2005 **Water drops on surfaces.**
Huamin Wang, Peter J. Mucha, and Greg Turk.
ACM Transactions on Graphics (SIGGRAPH), vol. 24, no. 3, pp. 921–929, 2005.
- Towards space-time light field rendering.**
Huamin Wang and Ruigang Yang.
ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D), pp. 125–132, 2005.

RESEARCH EXPERIENCE

- 2004–2009 **Research Assistant**, Georgia Institute of Technology
 Advisor: Greg Turk
- Fall 2007 **Research Intern**, Microsoft Research Asia
 Mentors: Kun Zhou and Baining Guo
- Summer 2007 **Research Intern**, Microsoft Research
 Mentor: Hugues Hoppe
- Summer 2006 **Research Intern**, Adobe Research
 Mentor: Gavin Miller
- 2004 **Research Assistant**, Stanford University
 Advisor: Ron Fedkiw

BOOK CHAPTERS AND THESES

- 2011 Ruigang Yang, **Huamin Wang**, and Cha Zhang. *Computational Photography: Methods and Applications*, chapter 18: Dynamic View Synthesis with an Array of Cameras. CRC Press, 2011.
- 2009 **Huamin Wang**. *Practical Water Animation using Physics and Image Based Methods*. PhD thesis, Georgia Institute of Technology, August 2009.

SELECTED PATENTS AND PATENT APPLICATIONS

- 2024 **Huamin Wang** and Chen Liu. Systems and methods for arranging clothing patterns, 2024. US12175597B1.
Huamin Wang and Chen Liu. Systems and methods for arranging and displaying clothing patterns, 2024. US12175620B1.
 Chen Liu, Huang Chen, Gaofeng He, and **Huamin Wang**. Methods and systems for personalized image generation, 2024. US20250106369A1.
- 2022 Chen Liu and **Huamin Wang**. Parallel computation methods and systems for multiplying symmetric matrices with vectors, 2022. US20250036716A1.
 Chen Liu and **Huamin Wang**. Machine learning-based methods, devices, and computer-readable storage media for measuring bending stiffness of fabrics, 2022. US20250045499A1.
- 2009 Hugues H. Hoppe, Yonatan Wexler, Eyal Ofek, and **Huamin Wang**. Factoring repeated content within and among images, 2009. EP2252971B1.
- 2008 Gavin S. P. Miller and **Huamin Wang**. System and method for simulating shallow water effects on arbitrary surfaces, 2008. US7921003B2.